

## Review

# Modeling methods for GenCo bidding strategy optimization in the liberalized electricity spot market—A state-of-the-art review

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## ABSTRACT

The electricity market has since 1980s been gradually evolving from a monopoly market into a liberalized one for encouraging competition and improving efficiency. This brings the opportunity for generation companies (GenCos) to make more profits while embracing more risks of not being dispatched. Therefore, it has become a core interest for the GenCos to develop optimal bidding strategies to maximize the profits and minimize the risks while participating in such a competitive market. The literature pertaining to this issue has grown rapidly in recent years, and many different modeling approaches, such as mathematical programming, game theory, and agent-based models, have been investigated under the liberalized market environment. Meanwhile, along with the increasing penetration of renewable energy, the electricity market is facing more complexity and stochasticity from both uncertain generation and dynamic demands. The intermittent and unsteady nature of these renewable power sources motivates the GenCos to further optimize their bidding strategy by considering the new constraints. This paper presents a comprehensive literature analysis on the state-of-the-art research of bidding strategy modeling methods.

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## 1. Introduction

In a traditional monopolistic or vertically integrated electricity market, power providers mainly aim to minimize the expected costs while maintaining an adequate security of supply [1]. Since 1980s, however, the electricity markets have been gradually evolving toward liberalized or deregulated structures, which are characterized by open competitive energy markets, unbundling electricity services, open access to the network, etc. To establish a competitive electricity market and improve its efficiency, the restructured market allows for gaming on the market power and tends to stimulate the emergence of new technologies [2,3]. The electricity spot market can be regarded as a market where the electricity can be sold or purchased at varying price and delivered either immediately or at a given moment. Participants have to make decisions independently under complicated situations with insufficient information about their rivals and various uncertainties in the market such as load variations, competitor's behavior, and power system contingencies.

The deregulated electricity market behaves more like an imperfect competition or oligopoly market due to the special characteristics of the actual electricity market, such as a limited number of suppliers, long construction periods of power plants, large capital investment sizes, transmission constraints, and transmission losses [4]. Typically, only a few dominating generation companies (GenCos) serve a given geographic region. In such an oligopolistic market, an individual GenCo has its market power, that is, it can affect and manipulate market price via its strategic bidding behavior [5]. This indicates an opportunity for the GenCos to increase their profits through strategic bidding. Therefore, it is possible for the GenCos to maximize their profits by optimizing the bidding strategy in the deregulated electricity spot market while minimizing the associated risks. Also, the electricity market is a complex dynamic system with complicated interactions among physical structures, market rules and all participants. As such, each participating agent faces risks and uncertainties while pursuing profit maximization [6,7].

Meanwhile, renewable energies such as wind, solar, and biomass are regarded as a key factor in tackling global climate change and energy shortage crisis [8]. For example, wind energy has globally experienced fast growth during the past decade [9,10]. Along with the ongoing worldwide utilization of wind power and its increasing penetration into electricity markets, more wind

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GenCos (WGenCos) will participate in the electricity market by presenting offers and committing the delivery of the agreed amount of wind energy at a given moment [11]. It is well-known that regulation power must be reserved to deal with the possible fast load variations and unforeseen problems with production capacity [12]. With a high penetration of such intermittent generations, however, the regulation power needed will further increase in that such generations are characterized by large variations in addition to the aforementioned load swings and outages of production capacity [13]. When bidding in the electricity markets, wind GenCos must pay for energy production deviations resulted from the prediction error, which can be as much as 10% of the total generator energy incomes [14]. The optimal bidding strategy, especially in the deregulated electricity markets with high penetration of intermittent renewable energy, must consider the cost paid for the energy imbalance in relation with the above-mentioned uncertainties and constraints.

Therefore, one critical issue faced by all GenCos participating in the electricity spot market is how to optimize their bidding strategies according to the limited information available to maximize their profits. This problem has attracted much attention from both the academic circle and the industry. In particular, numerous publications on modeling and analyzing GenCo bidding strategy in the competitive electricity market have emerged since 2000, and great progress has been witnessed. However, to the best of our knowledge, the most recent literature survey on GenCos bidding strategy was performed 10 years ago [5]. This paper thus aims to provide a comprehensive literature review on the state-of-the-art research on this topic. It should also be noted that several review papers related to electricity markets can be found in literature, but they focus on different perspectives and are briefly summarized in the following. Foley et al. review the electricity systems modeling techniques, discuss various key proprietary models for electricity systems available in the U.S. and Europe, and provide an information resource on the choice of model in investigating different aspects of the electricity system [15]. Anderson reviews the restructuring efforts in the U.S., Australia, the EU and several EU nations, as well as the responses of consumers in those markets [16]. Aggarwal et al. review the main methodologies used in forecasting the electricity price in the deregulated markets [17]. Mid-eksa and Kallbekken review the research on how climate change would impact the electricity markets [18]. Besides, the review of the optimal power generation dispatch problem in both regulated and deregulated electricity markets can be found in [19] and [20].

The remainder of the paper is organized as follows. Section 2 gives a general introduction to the structure and operations of the liberalized electricity market. Thereafter, the related literature is categorized according to their modeling algorithms in Section 3 and further summarized in Sections 4–7, respectively. Conclusive remarks as well as some possible directions for future research are finally presented in Section 8.

## 2. Liberalized market structures and management rules

Participating in electricity market implies presenting bids and committing the delivery of the agreed amount of energy at a given moment. Reliable delivery of electrical energy to load centers entails a continuous process of scheduling and adjusting electricity generation in response to constantly changing demand [7]. If the actual energy delivered by one generator is greater or smaller than what is committed, the GenCo will pay for additional cost of maintaining the balance between the generation and the load. In deregulated electricity markets, a GenCo is usually an entity owning generating facilities and participating in the market with the sole objective of maximizing its benefit [21]. Most evidently,

individual bidding strategies are of essence to the interactions where the participants' actions affect others' possible outcomes.

Bidding strategies should be developed based on the market structure, auction rules, and bidding protocols. A variety of electricity markets restructuring models have been proposed, which generally can be categorized into three types: (a) market pools (PoolCo), (b) bilateral contract (BC) markets, and (c) hybrid markets (HM) [22]. PoolCos are popular among the approaches to organizing electricity trading. Essentially, a PoolCo is a more centralized marketplace where an Independent System Operator (ISO) clears the market according to the bids from both the sellers and the buyers, operates and manages the entire system, and maintains its reliability. In a PoolCo, different GenCos compete not for specific customers but for the right to supply energy to the grid. A bilateral contract model defines a flexible market where the participants can specify and negotiate the terms and conditions of trading agreements independent from the ISO. The ISO mainly ensures enough transmission capacity and security. Generally, the market structure of a fully deregulated hybrid electricity market can be illustrated by Fig. 1. It may contain not only power pools and bilateral contracts but also ancillary services (AS) such as frequency and voltage controls, load following, energy imbalance, spinning reserve, and supplementary reserve and standby reserve [15]. Competing generators offer their electricity output to retailers in a wholesale electricity market; the retailers then re-price the electricity and take it to the retail market. Although the wholesale pricing used to be the exclusive domain of large retailing suppliers, the markets are increasingly opening up to large end-buyers as well. The participants can directly sign bilateral contracts with others and/or bid in the PoolCos. Also, a fully deregulated hybrid market should allow the end buyers to choose among different competing suppliers in the electricity retail market. It should be mentioned that this paper mainly focuses on the literature pertaining to the electricity wholesale market in view that the GenCos usually do not interact with the end consumers directly. Besides, Haas and Auer emphasize six prerequisites for effective competition in reformed wholesale electricity markets: (1) separation of the grid from generation and supply; (2) wholesale price deregulation; (3) sufficient transmission capacity for a competitive market and non-discriminating grid access; (4) excess generation capacity from many competing generators; (5) an equilibrium relationship between short-term spot markets and long-term financial instruments for marketers to manage spot-market price volatility; and (6) an essentially hands-off government policy that encompasses reduced oversight and privatization [23]. Moreover, some representative electricity wholesale markets formed in different geographical locations are provided in Table 1.

Generally, a bid may include several energy price segments together with the corresponding quantity of electricity. As the most common structure, the pool-based market is an auction center in which all competitive participants are required to submit quantity-price pairwise bids they commit to receive from or pay to the pool. As illustrated in Fig. 2, once the bidding period ends, an ISO ranks and then matches the selling offers with buying bids so that the buying bids of the highest price are matched with the selling offers of the lowest price [24]. When all the demands are met, the price of either the last accepted offer or the first rejected offer could be set as the market clearing price (MCP). The pricing mechanism for all the rest dispatched GenCos can be either uniform pricing (UP) or pay-as-bid (PAB). The UP auction indicates that all the winning suppliers are paid at the same MCP. The PAB auction means that each winning supplier bidder is paid at its bidding price of the committed amount of electricity.

Each different market structure has its own auction rules and bidding protocols. Various auction rules can be categorized into

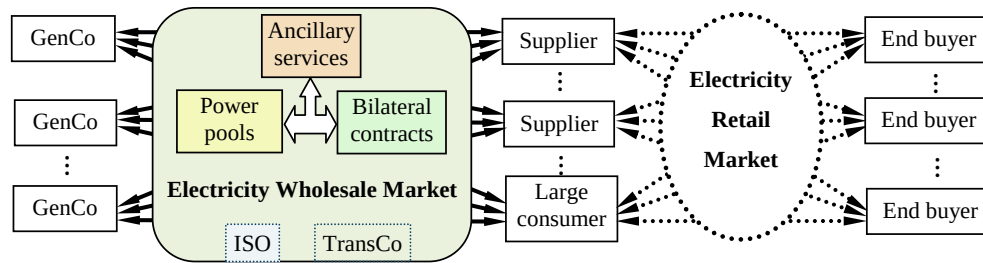


Fig. 1. The general market structure of deregulated electricity markets.

two types: static or dynamic. In static auctions, the bidders submit sealed bids, whereas dynamic auctions allow the bidders to observe others' bids so that they can revise their bids sequentially [5]. The static auction may work based on either the UP or PAB rule. Bidding strategy in PAB-based market is more complex and potentially more important than that in UP-based market. In this case, the GenCos should estimate the uncertain MCP and bid slightly less than it. The PAB rule represents the future trend in the deregulated electricity markets and is expected to lower the market prices and reduce the price volatility [25]. Although the majority of operating electricity markets currently still employ the sealed bid auction with the UP auction rule, extensive research has been carried out on the applications of PAB rule as well [26]. According to different market designs, the bidding protocols can be divided into two types: single-part bids or multipart bids [5]. Under the single-part bidding protocol, as adopted in the California type power exchange (PX) market, the GenCos bid independent

prices for each hour; the winning bid and the schedules for each hour are determined via a simple market clearing process according to the intersection of supply and demand bid curves. This decentralized approach does not require the ISO to make unit commitment decisions. Instead, the GenCos have to consider all involved costs and constraints in preparing their bids. Therefore, whenever a significant physical or technical constraint occurs in a generation unit, a modification mechanism should be applied to the schedule, e.g., via short-term balancing market. In contrast, a multipart bid, as addressed in the England–Wales or British type electricity market, may include separate prices for ramps, start-up costs, shut-down costs, no-load operations, and energy. Although this type of bidding protocol can reflect the cost structure and the technical constraints of generation units, the non-convex Unit Commitment (UC) problem might not converge to a global optimal solution for large scale systems, possibly resulting in inequitable dispatches for different GenCos.

**Table 1**  
Typical electricity wholesale markets in the world.

| Markets         | Introduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO New England | ISO New England, established in 1997, operates the region's day-ahead and real-time energy spot market with LMP, capacity market, forward reserves market, regulation market, and financial transmission rights market.                                                                                                                                                                                                                                                     |
| PJM             | Pennsylvania, New Jersey, and Maryland (PJM) market. It is a regional transmission organization that coordinates the wholesale electricity in 13 states and the District of Columbia; day-ahead and real-time spot market with LMP, regional and locational capacity market, financial transmission rights market, and bilateral contract market.                                                                                                                           |
| MISO            | Midwest ISO, established in 2002, administers the day-ahead and real-time energy market known as the Day-2 market with hourly LMP updating, and a monthly financial transmission rights (FTR) allocation and auction.                                                                                                                                                                                                                                                       |
| California ISO  | California ISO (CAISO), established in 1996, operates the region's wholesale electricity market including day-ahead, hour-ahead and real-time energy spot market with LMP, ancillary services, and financial transmission rights market.                                                                                                                                                                                                                                    |
| NYISO           | New York ISO, established in 1999, operates the day-ahead and real-time spot market with LMP, regional and locational capacity market, and financial transmission rights market.                                                                                                                                                                                                                                                                                            |
| ERCOT           | Electric Reliability Council of Texas (ERCOT), formed in 1970, operates both the transmission and the competitive energy market within the Texas interconnection as a single system. It operates the retail market including customer switching and the wholesale markets for balancing and ancillary service markets.                                                                                                                                                      |
| Dutch type APX  | Dutch Amsterdam Power Exchange (APX), established in 1999, allows the participants to electronically send before 10:30 their offers/bids for each hour between 00:00 and 24:00 of the next day; requires them to be responsible for the balance between the committed amount and the actual production.                                                                                                                                                                     |
| Nordic Elspot   | The spot market, Elspot at the Nordic power exchange (PX) Nord Pool, takes the form of a pool-based market where participants exchange power contracts for physical delivery in the following operation day.                                                                                                                                                                                                                                                                |
| APX-ENDEX       | APX-ENDEX (European Energy Derivatives Exchange), the successor of Dutch APX, is an Anglo-Dutch PX operating the spot and derivatives markets for electricity and natural gas in the Netherlands, the United Kingdom, and Belgium. APX-ENDEX provides exchange trading, central clearing and settlement, and data distribution services as well as benchmark data and industry indices.                                                                                     |
| Spanish PoolCo  | Spanish day-ahead spot market, introduced in 1998, works as follows. Before 11:00 a.m., qualified buyers and sellers present their offers for the 24 hourly periods of the following day. Offers can consist of up to 25 different price/quantity pairs for each period and for each generating unit they own (simple offer), or include indivisibility conditions, a minimum revenue condition, production capacity and scheduled stop conditions (complex offer).         |
| NEM Pool        | The National Electricity Market, under reconstruction since 1998, is an electricity wholesale market for Queensland, New South Wales, Victoria, South Australia, the Australian Capital Territory and Tasmania. Generators in the NEM bid their output in a wholesale pool where retailers and other participants buy the electricity at the wholesale spot price determined by NEMMCO based on the lowest cost generator bids required to meet the electricity demand.     |
| JEPX            | Japan Electric Power Exchange (JEPX), established in 2003, operates the wholesale electric power exchange and the related businesses.                                                                                                                                                                                                                                                                                                                                       |
| Ontario PoolCo  | Ontario electricity market, opened in 2002, consists of physical markets for energy and operating reserves as well as a financial market for transmission rights, administered by the Independent Electricity Market Operator (IEMO). The information available in advance to the participants is the forecasted demand and the predicted hourly Ontario energy price, while information related to historical bidding or hourly dispatch of competitors is not accessible. |

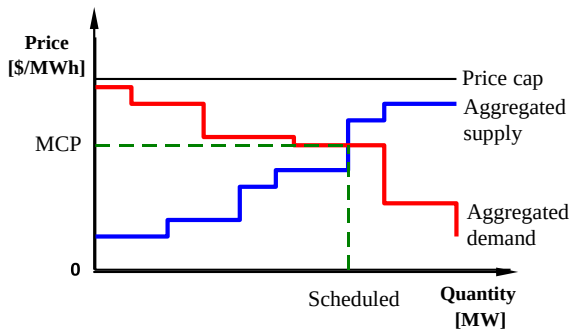


Fig. 2. Market clearing mechanism.

### 3. Modeling methods for bidding electricity in spot markets

In 1988, Schweppe noticed that some electricity utilities were moving away from rigid fixed prices toward a spot price marketplace, and he discussed the management of four aspects of such new price structures: the electricity marketplace itself, the customer, the utility, and the regulatory commission [27]. Later, David formally addressed the strategic bidding issue for competitive power suppliers and developed a conceptual optimal bidding model and a dynamic programming method for England–Wales type electricity markets, in which each GenCo bids a fixed price for each generation block [28]. Since then, the strategic bidding problem for competitive electricity supplier has attracted more attention, and various modeling approaches have been proposed to generate strategic bidding strategies [29].

We classify the various modeling literature for bidding strategy analysis in the electricity spot market into four groups: (1) single GenCo optimization models, (2) game theory based models, (3) agent-based models, and (4) hybrid or other models, as illustrated in Fig. 3. Each group of models may be further divided into small subgroups according to the model formulation and solution algorithms. For example, the single GenCo optimization models include many mathematical programming methods such as Mixed Integer Programming (MIP), Nonlinear Programming (NLP), and Dynamic Programming (DP); the game theory models might adopt different competition rules: Bertrand competition, Cournot competition, Supply Function Equilibrium (SFE), and some other newly proposed competition rules; the agent-based models can be categorized in terms of different learning algorithms such as model-based adaptation algorithms (MA), genetic algorithms (GA), Q-Learning (QL), computational learning (CL), Ant Colony Optimization (ACO), etc. Generally, the single GenCo optimization models focus on only one specific player while simplifying the rest players and the influencing factors as a set of deterministic or stochastic independent

variables, while game theory and agent-based approaches deal with the situation of more than one player in the market. Game theory equilibrium models investigate the bidding strategies from the perspective of players' mutual interactions. Agent-based models tend to mimic human behaviors and simulate optimal bidding strategies [30]. The main characteristics of the three modeling approaches are provided in Table 2. Recently, some hybrid and non-conventional methods have also been proposed. The representative publications in the four groups are reviewed and summarized in the following sections, respectively.

### 4. Single GenCo optimization models

In earlier publications, the issue of optimal bidding strategy selection is often addressed as a cost-minimization problem and solved via traditional cost-based UC algorithms [31]. More recently, under the assumption that the MCP could be regarded as an exogenous variable [32], many mathematical programming approaches have been applied to address the problem of optimal bidding strategy selection. The majority of model formulations incorporate stochastic probabilistic elements, either in the problem data (e.g., the objective function and the constraints), or in the algorithm (through random parameter values, random choices, etc.), or in both [33]. An insightful discussion on the application of stochastic programming methods to the energy market can be found in [34]. The typical optimization methods adopted in the literature include Integer Linear Programming (ILP), Mixed Integer Programming (MIP), Multi-Objective Linear Programming (MOLP), Nonlinear Programming (NLP), Dynamic Programming (DP) [15], newsvendor, and Markov decision process (MDP) models. As mentioned before, the literature adopting these models typically optimizes the bidding strategy for a single-market participant while ignoring or simplifying the behavior aspects of other players. The representative references are summarized in Table 3, and more discussion is provided in the following.

As reviewed in [35], many mathematical programming problems in a competitive electric energy framework can be modeled as mixed-integer linear programming (MILP) models. In [36], an MILP model is formulated for a price-maker GenCo to solve the self-scheduling problem and maximize the profit in a pool-based electricity market. Conejo et al. propose several mathematical programming models for a price-taking thermal GenCo to derive the optimal bidding strategy in a pool-based market with highly uncertain MCPs. The problem is first modeled as a stochastic mixed-integer linear programming (SILP) model and then transformed to two MILP models, one of which could be easily solved using a commercial ILP solver. Then a simple but informative bidding rule is derived from the solution [37]. Similarly, from the perspective of a price-taking hydropower GenCo (HGenCo)

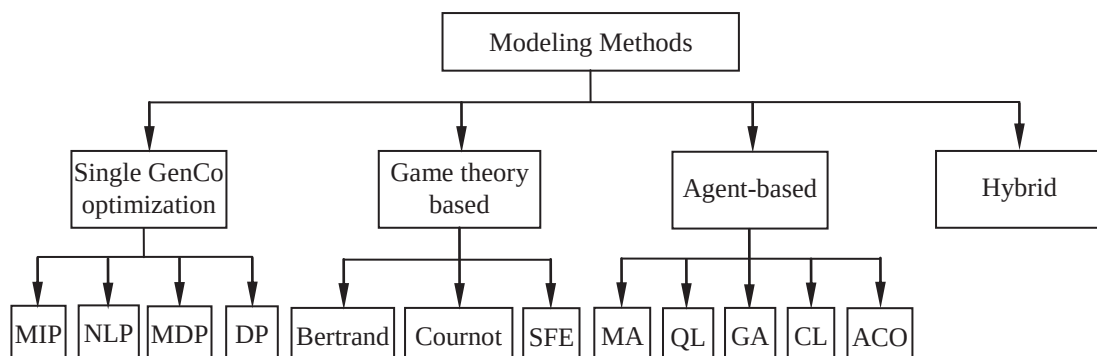


Fig. 3. Modeling methods for bidding electricity in the spot market.

**Table 2**  
Characteristics of three types of models.

| Models                    | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Single GenCo optimization | Developing optimization models to describe the entities in the electricity market with the objective of finding an optimal solution: <ul style="list-style-type: none"> <li>• Well-established and solid mathematical foundation</li> <li>• Generally focusing on one specific player in the system by simplifying the rest of the system as a set of exogenous variables</li> <li>• Usually modeling no aspects of players' intelligent behaviors</li> <li>• Difficult to model the complex, uncertain and dynamic systems or analytically derive the optimal bidding strategy for the GenCos in the deregulated electricity markets</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Game theory               | Modeling the electricity market as a game and mathematically capturing the players' behavior in the game where one player's success in making choices depends on the others' choices <ul style="list-style-type: none"> <li>• Usually mathematically well-defined, involving a set of game players, a set of bidding strategies, and a specification of payoffs for each possible combination of bidding strategies</li> <li>• Analyzing the economic equilibria of the electricity market by focusing on the players' interactions</li> <li>• Capable of providing analytical rationale and explanation on how strategic bidding behaviors affect the GenCos' market power and profits</li> <li>• All players are assumed to be rational, which does not generally hold in the reality</li> <li>• The frustrating issue of multiple equilibria often occurs in solving the model</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Agent-based               | Modeling the complex electricity market as collections of rule-based agents interacting with one another dynamically and intelligently, simulating human beings' behavior to make optimal bidding strategies <ul style="list-style-type: none"> <li>• Only a few simple rules are specified for and followed by various agents that situated in the network and behave intelligently in the system</li> <li>• Agents usually have and only require imperfect, local information and visibility</li> <li>• No centralized control or planning is required although random elements often exist either among variable agents or in the system</li> <li>• Agents can interact with each other directly or through the environment, resulting in complex emergent global behavior of dynamic-equilibrium and adaptation</li> <li>• More flexible, robust, and easily implemented compared with analytical approaches</li> <li>• Capable of capturing the details about agents behaviors, which is helpful in figuring out the relationships between individual decisions and system behavior</li> <li>• Capable of modeling the dynamics of systems that are not in equilibrium as well</li> <li>• The underlying mathematical foundation is still not well developed</li> <li>• Requiring computation-intensive procedures</li> </ul> |

participating in the PX of Nord Pool, a day-ahead power market, Fleten and Kristoffersen transform a two-stage SILP model to an MILP model for determining optimal bidding strategies by taking into account the discrete market price scenarios. The effect of market price uncertainty on bidding optimization is explicitly explored by comparing the stochastic approach to the deterministic counterpart [38]. Angarita et al. present the application of stochastic optimization technique in maximizing the joint profit of hydro and wind generators in a pool-based electricity market [39]. To handle the uncertainty of wind prediction, the hourly wind power output is considered as a discrete random variable in the optimization problem. Compared to other possible bid strategies that make use of the expected wind power value, the combined bidding strategy obtains significant improvements. Also, it is a useful tool for GenCos to avoid penalty costs or income reduction. Sen et al. propose a multi-stage SILP model for scheduling and hedging in wholesale electricity markets [40]. This SILP model captures stochastic electricity demand, electricity forward price, gas forward price, and spot price of electricity. Based on the structure of the SILP model, a nested column generation decomposition strategy is proposed to decompose the model into three interrelated sub-problems. The experimental results demonstrate that the proposed approach could provide robust decisions for scheduling and hedging problems. Besides, Ni et al. develop a stochastic mixed-integer program (SMIP) to systematically handle the MCP uncertainties, bidding risk management, and self-scheduling requirements for a hydrothermal GenCo to maximize profits under a deregulated market. The model is solved by the proposed algorithm combining Lagrangian relaxation and stochastic dynamic programming (SDP) method [41].

De Ladurantaye et al. introduce an SILP model to maximize the profits for a hydropower GenCo of multiple power plants along a river in a deregulated market. The proposed model aims to support the price-taking GenCo in its day-ahead bidding decisions [42]. Morales et al. address a multi-stage SILP problem which determines the best bidding strategy for a WGenCo in an electricity

market including various trading floors (e.g., day-ahead market and a sequence of adjustment markets). In the SILP problem, four uncertain sources are considered: wind power generation, day-ahead market price, adjustment market price and imbalance energy price. The multi-stage SILP problem is formulated as a linear programming (LP) model, and a case study for a WGenCo in Kansas is conducted for the illustration of solving the LP model [43].

Gross and Finlay develop a nonlinear programming (NLP) model to optimize strategic bids of a GenCo in a multi-period auction market under the assumption of perfect competition, and propose a Lagrangian relaxation (LR) method to solve the NLP model [44]. Similarly, to deal with the GenCo's bidding optimization and self-scheduling problem, Zhang et al. develop an NLP model by considering uncertain bidding information of other participants, and solve this model by an LR method [45]. Yucekaya et al. present two particle swarm optimization (PSO) algorithms to determine bid prices and quantities under the rules of the Pennsylvania, New Jersey, and Maryland (PJM) market. The first one employs a conventional PSO technique whereas the second uses a decomposition technique in conjunction with the PSO approach. It is found that the latter algorithm can dramatically outperform the former. Also, it is shown that for nonlinear cost functions, PSO solutions provide higher expected profits than marginal cost-based bidding [22]. Similarly, while considering the non-convex operating cost functions of thermal generating units and minimum up/down time constraints, Boonchuay et al. propose an optimal risky bidding strategy for a GenCo by self-organizing hierarchical particle swarm optimization with time-varying acceleration coefficients (SPSO-TVAC) [46]. With rivals' behavior in competitive environment being simulated via the Monte Carlo method, the significant risk index based on mean–standard deviation ratio (MSR) is maximized to generate the optimal bid. The proposed SPSO–TVAC approach is concluded to be capable of providing a higher MSR than other PSO methods.

Wen and David propose a stochastic nonlinear programming (SNLP) model for deciding optimal bidding strategies for

**Table 3**

Representative optimization based modeling methods for bidding strategy analysis.

| Ref.                       | Models/Methods    | Auction rules | Markets           | Assumptions                                                                 | Applications                                                                    |
|----------------------------|-------------------|---------------|-------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| de la Torre et al. [36]    | MILP              | UP            | PoolCo            | Price-maker GenCo                                                           | Self-scheduling in a PoolCo; market power analysis                              |
| Conejo et al. [37]         | SILP/MILP         | UP            | PoolCo            | Price-taking GenCo; uncertain MCP                                           | A probabilistic framework for treating uncertain MCP                            |
| Fleten et al. [38]         | SILP/MILP         | UP            | Nord Pool         | Price-taking GenCo; uncertain MCP                                           | Exploring the effects of the uncertain market price in day-ahead markets (DAMs) |
| Angarita et al. [39]       | SILP/MILP         | UP            | Spanish type PX   | Price-taking WGenCo and HGenCo                                              | Combined strategy for bidding and operating in DAMs.                            |
| Sen et al. [40]            | Multi-stage SILP  | UP            | Forwards and Spot | Monthly forward contracts; forward prices reflect the expected spot prices  | Power portfolio optimization in the wholesale market                            |
| Ni et al. [41]             | SMIP              | UP            | ISO New England   | Power blocks can be different for different hours; Markovian MCPs           | Maximizing the profit and managing bidding risks in DAMs                        |
| De Ladurantaye et al. [42] | SILP              | UP            | General DAM       | Price-taking GenCo; known MCPs                                              | Hydropower GenCos' profit maximization in DAMs                                  |
| Morales et al. [43]        | SILP/MILP         | UP            | Hybrid            | Rational bidders; risk-neutral case; no penalty for balancing market bidder | Decreasing WGenCos' risk of profit variability                                  |
| Gross et al. [44]          | NLP               | UP            | British type PX   | Perfect competition for GenCos; individual bids do not affect MCP           | Multi-period auction rule                                                       |
| Zhang et al. [45]          | NLP               | UP            | ISO New England   | Quadratic bid function with rivals' being available as distributions        | Cost minimization of bidding in DAMs                                            |
| Yucekaya et al. [22]       | NLP/PSO           | UP            | PJM               | The uncertain MCP is exogenous to the bidding decisions                     | GenCo bidding for maximum profits                                               |
| Boonchuay et al. [46]      | NLP/PSO           | UP            | PoolCo            | Step-wise bidding protocol                                                  | Risk management of profit variation of GenCo in the pool                        |
| Wen et al. [4,47]          | SNLP/GA           | UP            | California ISO    | Sealed bids on 24 linear supply functions w/o price caps                    | Market power analysis for DAMs                                                  |
| Ma et al. [48]             | SNLP              | UP            | PoolCo            | Single buyer; sealed bid on linear supply function with power caps          | Minimizing GenCos' risks while maximizing their profits.                        |
| Guan et al. [49]           | NLP               | UP            | California ISO    | Bid can be submitted for individual unit or as an aggregated bid            | Providing "good enough" bids in DAMs with less computation                      |
| Usaola et al. [50]         | SLP               | UP            | Spanish PoolCo    | Zero bid price of WGenCo; perfect price forecast                            | WGenCo bidding for maximum profits                                              |
| Zhang [51]                 | Statistical model | UP            | NYISO             | Perfectly inelastic and exogenously given load profile                      | Demand side management                                                          |
| Rahimiyan et al. [52]      | News-vendor       | PAB           | General DAM       | Normally distributed MCPs                                                   | Risk analysis and profit maximization in a PAB PoolCo for DAMs                  |
| Song et al. [53]           | MDP               | UP            | General DAM       | Operational constraints & demand-side bid are ignored; risk-neutral GenCo   | Optimizing bidding decisions over a planning horizon                            |
| Gajjar et al. [54]         | MDP               | UP            | PoolCo            | Known forecast daily load with load variation being ignored                 | Long-term profit maximization with stochastic risks                             |
| Bathurst et al. [55]       | MDP               | UP            | Hybrid            | Different imbalance price assumptions                                       | WGenCo imbalance cost minimization in the hybrid                                |
| Pinson et al. [56,57]      | News-vendor       | UP            | Dutch type APX    | Price-taking WGenCo; no internal balancing control; selling all for MCP     | Revenue maximization of WGenCo in DAMs                                          |

competitive power suppliers in a sealed bid auction-based electricity market. The model assumes that the power supplier bids a linear supply function and is paid at the MCP with the system dispatch levels being stipulated by a market operator to minimize customers' payments. It is shown that the MCP can be significantly higher than the competitive levels if the suppliers bid strategically, and that the market power of the suppliers will be reduced if the load is elastic to the price of electricity [4]. Similarly, they build more SNLP models and propose a genetic algorithm based method to build bidding strategies for power suppliers in the California-type day-ahead energy market in which power suppliers are required to simultaneously bid 24 linear energy supply functions, one for each hour, and the system dispatch levels are stipulated separately for each hour by utilizing the UP pricing rule. The method is believed to be especially suitable for those suppliers with marginal or near-marginal generating units [47]. Ma et al. develop an SNLP model for optimizing bidding strategies by considering the risks for the GenCos participating in a pool-based single-buyer electricity market. Each GenCo is assumed to bid a linear supply function and the system is dispatched to minimize the total purchasing cost of the single-buyer. Each GenCo chooses the coefficients in the linear supply function for making tradeoff between

two conflicting objectives: profit maximization and risk minimization [48]. Guan et al. develop a bidding strategy based on the theory of ordinal optimization. The basic idea is using an approximate model to describe the influence of bidding strategies on the MCP. A nominal bid curve is obtained by solving optimal power generation for a given set of MCPs via Lagrangian relaxation. The best bid is then selected by solving full hydrothermal scheduling or unit commitment problems [49].

Usaola et al. formulate a stochastic linear programming model by including the probability density of wind forecasting and analyze the optimal bidding strategy for a WGenCo in a spot market. It is found that the most accurate prediction is achieved when bids are updated in intraday markets by using more recent predictions. However, the most accurate prediction cannot ensure the highest revenues due to the different prices of spilled and bought energy and the bias of the prediction programs. In order to generate maximum revenue, the uncertainty of the power prediction must be considered [50]. Zhang uses a dynamic random effect ordered probit model to analyze the GenCo's bidding behavior in the NYISO day-ahead wholesale market. The results show that generators in higher-priced groups tend to withhold their capacity strategically to push up market prices. It is also verified that

demand conditions might affect market prices significantly [51]. Rahimiyan et al. formulate the bidding decision-making problem from a supplier's viewpoint in a PAB auction spot market by assuming a normally distributed MCP, and analyze the effect of some risk factors on the supplier-expected benefit and selling amount [52]. Song et al. propose a bidding decision-making strategy in which the impacts of production limit and market share on the optimal bidding strategies are considered. It is concluded that the Markov decision process (MDP) model is able to optimize the decision over a planning horizon. However, the model makes a few strong assumptions such as ignoring the power system operational constraints and giving no provision for incorporating risk attitude in an ordinary MDP [53]. Gajjar et al. formulate the GenCo optimal bidding in a deregulated power market in the framework of MDP [54]. An optimal strategy is devised to maximize the profit by employing the temporal difference technique and actor-critic learning algorithm. The method is concluded to be especially useful for long-term profit maximization under stochastic risks. Bathurst et al. present a strategy for bidding a few hours before the operation time for the wind producers under the New Electricity Trading Arrangements (NETA), changed to British Electricity Trading Transmission Arrangements since 2005, by using Markov processes for simulating a wind farm [55]. The method demonstrates substantial reductions in the imbalance costs as well as the effect of market closure delays and window lengths of wind forecasting. Pinson et al. study daily bidding by using the rules of the Dutch APX electricity market, where bids are presented only once per day and not updated in intraday markets [56]. It is found that as a result of reduced regulating market costs from better hourly predictions to the market, wind power producer could obtain up to 8% more profit if the time between market bids and delivery is shortened from the day-ahead Elspot market. They also formulate a general methodology for deriving optimal bidding strategies based on probabilistic forecasts of wind generation in the form of predictive distribution. This flexible methodology can be tailored based on the needs of a specific market participant [57].

Besides, bi-level optimization is often applied either to represent the strategic interaction among suppliers or in hybrid markets where electrical energy and spinning reserve are simultaneously traded [58,59] or in the presence of future contracts [60] and bilateral contracts [61]. Also, the influence of extra objectives such as the minimization of supplier emission of pollutants [62], or the influence of unit reliability [63] has been analyzed by using optimization models. The competition process can be represented as a dynamic feedback system as well [2]. Attaviriyapap et al. propose an algorithm for determining the optimal bidding strategy for a GenCo in the deregulated day-ahead power and reserve markets [64]. The optimal bidding parameters for both markets are determined by solving an optimization problem, which considers unit commitment constraints such as generating limits and unit minimum up/down time constraints. In the study, evolutionary programming (EM) technique is used to solve the problem. In addition, a few other single GenCo optimization studies on bidding strategy in the electricity spot market can be also found in literature [65–72].

## 5. Game theory models

Game theory models, also called equilibrium models, optimize the bidding strategies by investigating players' interactions and analyzing economic equilibria of the system. Typically, in a game, each player chooses the strategy from its own strategy set; then a payoff will be assigned to each player by the payoff function; as a result the optimal solution can be reached via Nash equilibrium. Nash equilibrium is a strategy combination of all players in which

no player can increase its payoff by changing its own strategy alone so that every player will finally choose its strategy exactly as the equilibrium strategy combination. Game theory models provide analytical rationale and explanation on how market power can be exercised via strategic bidding behavior, but the assumption that all players are rational usually does not hold in practice [30]. Also, it is limited by the requirement of common knowledge on all GenCos' actual production costs [29].

One major criterion for classifying game theory based methods is the level of competition: cooperative and non-cooperative [73]. According to the competition level in the liberalized electricity market, three general types of game models in imperfect competition, namely, Bertrand, Cournot, and SFE, have been proposed in the recent literature. As the most competitive model, a Bertrand competition model is an oligopolistic framework where the GenCos compete with one another by using prices as the strategic variables and ignore their capacity constraints. In classic Cournot models, however, the GenCos compete by using quantities as strategy choices, under the assumptions of homogenous products, price-responsive demand, and an MCP is determined by the intersection of aggregated supply and market demand curves. In the SFE models, the GenCos compete through the simultaneous choice of supply functions [63]. The competition level as well as the derived price equilibria generally lies between the Bertrand and Cournot model [74]. Recently, some other methods are also used to model and analyze the strategic behavior in deregulated electricity markets.

### 5.1. Bertrand competition based models

In a standard Bertrand competition model, identical sellers are assumed to have constant unit costs and no capacity constraints in competing on the price offers to consumers, and this will inexorably cause the identical sellers to price at marginal cost [75]. However, the electricity spot market deviates from the standard Bertrand price game. It is a capacity-constrained oligopoly and the marginal cost pricing is unlikely to be an optimal bidding strategy [76]. Therefore, only a few relevant references can be found in recent publications. The representative Bertrand competition based models are summarized in Table 4.

Federico and Rahman analyze the effects of changing auction rule from UP in the wholesale market to PAB under two polar market structures (i.e., a perfect competition or Bertrand structure and a perfect collusion or monopoly bidding) with demand uncertainty [77]. It is found that under Bertrand structure there is a tradeoff between efficiency and consumer surplus while changing to the PAB rule. Also, a move from UP to PAB under monopoly conditions has a negative impact on profits and output (weakly), a positive impact on consumer surplus, and ambiguous implications for welfare and average prices [77]. Based on generalized Bertrand game, Ernst et al. propose to optimize the profit and obtain a strategic bid by assuming that each GenCo has a constant marginal cost over its domain of generation and its rivals do not change their bids from the last round of market to the next one [78]. They further adopt quadratic cost functions for GenCos and include a supply function in the bids instead of playing a Bertrand game [79]. Hu et al. define Bertrand-Edgeworth (B-E) auctions as a modified version of Bertrand-Edgeworth games where the demand is inelastic and a price cap is set exogenously [80]. B-E auctions are motivated by the discriminatory procurement auctions used in some wholesale electricity markets. They characterize the equilibrium structure for B-E auctions with multiple asymmetric bidding suppliers. Based on a proposed numerical algorithm, it is numerically illustrated that a weak (low-capacity) bidder does not necessarily price more aggressively in an oligopoly market [80].

**Table 4**  
Representative Bertrand competition based models for bidding strategy analysis.

| Ref.                 | Model bases        | Auction rules | Markets                  | Assumptions                                                                                    | Applications                                             |
|----------------------|--------------------|---------------|--------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Federico et al. [77] | Bertrand           | PAB           | British type PoolCo      | Polar structure— perfect competition or collusion                                              | Market design under the PAB auction rule                 |
| Ernst et al. [78]    | Bertrand           | UP            | Simplified 2-node PoolCo | The same bid posted as the previous MCP by other agents; known transmission capacity and loads | GenCo's profit maximization and power market design      |
| Minoia et al. [79]   | Bertrand           | UP            | Simplified 2-node PoolCo | Each session of the market is regarded as a stage                                              | GenCo's profit maximization and market design            |
| Hu et al. [80]       | Bertrand-Edgeworth | PAB           | Simplified               | Inelastic demand and exogenous price cap                                                       | Profit maximization in the dynamic oligopolistic market  |
| Bunn et al. [81]     | Bertrand           | PAB           | E&W PoolCo               | NETA-based; transmission limits ignored; GenCos independent of suppliers                       | Analysis of market power and market design policy issues |

Bunn et al. [81] develop a stylized Bertrand game with constraints to explore the main strategic decisions faced by the GenCos in the England and Wales (E&W) market based on the proposed NETA.

### 5.2. Cournot competition based models

Compared with Bertrand price-setting strategies, the quantity-setting equilibrium is more realistic for the electricity market. The Bertrand equilibrium assumes that by providing a lower price than others, a firm can capture the entire market demand and then meet it by expanding its output. However, such assumption is not tenable in view of the increasing marginal cost of electricity generation at a time point and the generation capacity constraints. Because the GenCos provide a homogeneous product, the Cournot assumption that firms make strategic decisions by quantity-setting behavior is considered to be a better approximation to reality than the price-setting assumption [82,83]. The representative Cournot competition based models are summarized in Table 5.

Borenstein and Bushnell use demand and plant-level cost data to simulate the competition in a restructured California electricity market [84]. This approach recognizes that firms might have an incentive to restrict output in order to raise price, and it enables the explicit analysis on each firm's ability to do so. It is found that, while the results make the deregulation of generation less attractive than where there is no market power, they do not suggest that deregulation is a mistake. It is argued that policies promoting the responsiveness of both consumers and producers to price fluctuations can significantly affect the reduction of market power. Willems studies a Cournot competition based game with two GenCos who share one transmission line with a limited capacity to supply price-taking consumers [85]. In the model, three different rules for the network operator to allocate transmission capacity are investigated, namely, all-or-nothing, proportional, and efficient rationing. Their effects on the GenCos competition and revenue are

analyzed. Kian et al. use feedback Nash-Cournot strategies for the market participants to bid in dynamic electricity double-sided auctions [86]. The simulation results show that compared with single-sided auctions, double-sided auctions are more efficient and lead to more stable and competitive MCPs. Tamaschke et al. focus on measuring the extent to which market power has been exercised in a deregulated electricity generation sector, and emphasize the need to consider the concept of market power in a long-term dynamic context [87]. A market power index is constructed by considering the differences between the actual market returns and long-term competitive returns, and it is estimated by using a mathematical optimization model. The results suggest that generators have exercised significant market power from deregulation. To investigate whether individual participants can increase profits by withholding generation from the market, Ahn and Niemeyer develop a Cournot-based model of Korean power system for a set of loads representing the load duration curve for Korea's system loads in 2002 [88]. The results indicate a strong possibility for the exercise of market power to increase market price in Korean market.

Krause et al. perform a Nash equilibrium analysis by defining a pool market as a repeatedly played matrix game and compare it with an agent-based model. It is concluded that GenCos may act strategically by bidding above their marginal production costs [3]. Kang et al. also propose a bidding model by using a two-player static game theory and analyze both demand-fixed and demand-under-uncertainty scenarios. The objective is to exploit a methodology to build the optimal bidding strategies for competitive power suppliers in day-ahead auction-based electricity markets with only the information on their possible future profits estimated from forecast system demand. It is indicated that a precise forecast of the demand can help the player to gain the advantage in the game [21]. Under the Cournot assumption, Park et al. analyze the power transactions in a deregulated energy marketplace such as PoolCo by

**Table 5**  
Representative Cournot competition based models for bidding strategy analysis.

| Ref.                   | Model bases    | Auction rules | Markets           | Assumptions                                                                                                                       | Applications                                                    |
|------------------------|----------------|---------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Borenstein et al. [84] | Static Cournot | UP            | Early California  | Various assumptions for bidders in multiple scenarios                                                                             | Comprehensive market analysis and new market design             |
| Willems [85]           | Cournot        | UP            | Simplified market | 2 GenCos; 1 TransCo; 3 allocating rules                                                                                           | Market design                                                   |
| Kian et al. [86]       | Cournot        | UP            | Simplified market | Double-side auctions; no limit to price area                                                                                      | Profit maximization in the dynamic oligopolistic market         |
| Tamaschke et al. [87]  | Cournot        | UP            | NEM Pool          | Present value of the generation capital is assumed to be repaid over the useful plant life; several ancillary costs are included. | Market power analysis in both short-run and long-run contexts   |
| Ahn et al. [88]        | Cournot        | UP            | KEPCO             | Varying electricity inputs; with/without withholding                                                                              | Market power analysis                                           |
| Krause et al. [3]      | Cournot        | UP            | PoolCo            | Constant loads; linear GenCo marginal costs                                                                                       | Assessment of the market dynamics of network-constrained PoolCo |
| Kang et al. [21]       | Cournot        | UP            | PoolCo            | Only 2 GenCos, knowing each other's payoffs and strategies                                                                        | Profit maximization                                             |
| Park et al. [89]       | Cournot        | UP            | PoolCo            | 2-player game with complete information                                                                                           | Profit maximization and market analysis                         |

modeling it as a non-cooperative game with complete information and determining the solution in a continuous strategy domain. A new hybrid solution approach employing a two-dimensional graphical approach as well as an analytical method is proposed to provide more apprehensible analysis [89].

### 5.3. SFE based modeling methods

The Supply Function Equilibrium (SFE), originally introduced by Klemperer and Meyer [90], is a way of describing how competitors could maximize profits in the competitive market of a single product with uncertain demands. In such a market, the participants prefer to set supply functions rather than compete in prices (Bertrand competition) or quantities (Cournot competition) [90]. Green and Newbery further advance the SFE theory by considering capacity constraints in analyzing the competition in the British electricity spot market, and they develop a model for the market of privatization [91]. It is verified that supply curve bidding (SCB) can better benefit the GenCos compared to fixed quantity-price bids. The advantages of SFE models compared with other models are also discussed [92]. The SFE model, which constitutes a good compromise between the Cournot and Bertrand models, is believed to most accurately reflect the actual behavior of players in the real power markets. Also, it is more appropriate for the centralized markets where each GenCo bids in terms of a supply curve [6]. Therefore, the worldwide ongoing deregulation of the electricity market has been stimulating the SFE based modeling analysis of strategic bidding behavior. The representative SFE based models are summarized in Table 6, followed by further discussion.

Li and Shahidehpour propose an SFE based modeling method for analyzing the competition and bidding strategy among GenCos with incomplete information while taking into account transmission constraints [93]. The competition is modeled as a bi-level problem in which the upper subproblem maximizes individual GenCos' payoffs and the lower subproblem clears the market. Sensitivity functions are developed for each GenCo's payoff with respect to its bidding strategies in order to solve the bi-level

problem. An eight-bus system is employed to illustrate the proposed method, and the numerical results show the impact of transfer capability on GenCos' bidding strategies. In the Standard Market Design (SMD) setup for electricity markets, Al-Agtash presents an SCB approach that iteratively alters the SFE model solutions and selects the best bid based on both the market-clearing LMP and network conditions [94]. This enables the GenCos to derive their best offering strategy both in the DAM and the long-term contractual markets. However, the results could vary significantly from one system to another according to the system characteristics such as network topology and the associated generation capacities of suppliers [94]. Other general SFE based modeling studies for strategic bidding in the electricity market can be found in literature [95–98].

It is observed that the SFE models are difficult in embracing congestion conditions, capacity constraints, and large systems with significant number of generators, unless strong restrictions are placed [99]. The supply curve bids obtained from the SFE models may not lead to a maximum profit, especially when the network is highly constrained. In view of this, Alaghtash and Yamin reformulate the supply function equilibrium model for a GenCo owning a number of generators and present a new approach for optimal supply curve bidding (OSCB) using Benders decomposition [99]. In their model, the offers of individual generators are simultaneously optimized to maximize the GenCo's total profit by taking into account physical constraints as well as transmission security constraints.

Many existing applications found in literature are limited to small systems due to the difficulty of analytically calculating the SFE for large systems. In view of this, Bompard et al. present an analytical approach to represent the strategic bidding behavior of the GenCos using an SFE model for large systems, in which the decision variables of the GenCos are the slopes of the supply function [100]. The proposed approach proves to be rather precise in determining the SFE and it can consider the GenCos' capacity limits. It is also shown that generation capacity constraints may contribute to GenCos' market power. One GenCo may be pivotal if

**Table 6**  
Representative SFE based models for bidding strategy analysis.

| Ref.                          | Model bases | Auction rules | Markets              | Assumptions                                                                              | Applications                                                                                               |
|-------------------------------|-------------|---------------|----------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Klemperer et al. [90]         | SFE         | UP            | Simplified market    | GenCo's cost curves are independent of SF choices; Various other assumptions             | Market power analysis; market design                                                                       |
| Green et al. [91]             | SFE         | UP            | Early British PoolCo | Smooth supply schedule with GenCo capacity limits                                        | Market power analysis with capacity constraints                                                            |
| Hobbs et al. [95]             | SFE         | UP            | Simplified PoolCo    | Slope-fixed linear SF curve; unchanged rival bids                                        | Bidding optimization with transmission constraints                                                         |
| Niu [96]                      | SFE         | UP            | ERCOT                | Homogeneous and divisible generations; no transaction cost.                              | Market efficiency analysis and market power analysis                                                       |
| Sioshansi et al. [97]         | SFE         | UP            | ERCOT BES            | Dispatched Generations; constant marginal cost                                           | Bidding under feasibility constraints in the balancing electricity service (BES) market; BES market design |
| Haghighat et al. [98]         | SFE         | PAB/UP        | PoolCo               | Slope-fixed linear affine SF curve; unchanged rival bids                                 | Bidding in the PAB/UP market; pricing rule analysis                                                        |
| Alaghtash et al. [99]         | SFE         | UP            | General PX (DAM)     | Constrained transmission; single-side bidding                                            | Solving OSCB problem for GenCos owning many generators                                                     |
| Bompard et al. [100]          | SFE         | UP            | General DAM          | Bid on the slope of the linear supply function                                           | Bidding with generation capacity and network constraints for DAMs                                          |
| Genc et al. [101]             | SFE         | UP            | General DAM          | GenCos of symmetric SFE and enough capacity                                              | Market power analysis with capacity constraints for DAMs                                                   |
| Holmberg [102]                | SFE         | UP            | General DAM          | Asymmetric GenCos with capacity constraints                                              | Market power analysis with capacity constraints for DAMs                                                   |
| Vahidinasab et al. [6]        | SFE         | UP            | PoolCo               | Single-side auction; quadratic bidding curve                                             | Impacts of GenCo's pollutant emission on bidding strategies                                                |
| Gao et al. [30]               | SFE         | UP            | General DAM & Fuel   | Single-sided auction; linear demand function                                             | Multi-period and multi-market bidding optimization; transmission planning                                  |
| Sahraei-Ardakani et al. [103] | SFE         | UP            | PoolCo               | Rational GenCos; slope-fixed linear affine SF curve; unchanged rival bids; fixed demands | Bidding under many realistic constraints; market power analysis                                            |

its rivals are capacity constrained. It could substantially raise the market price by unilaterally withholding the output.

To fully consider the impact of capacity constraints and pivotal firms on equilibrium predictions, Genc and Reynolds characterize the set of symmetric SFEs for capacity-constrained GenCos in the UP-auction wholesale market. It is shown that the rise of GenCos' capacities could lead to the increase of this set of equilibria [101]. Holmberg also numerically studies the asymmetric supply function equilibrium with capacity constraints and shows that in this case, the valid SFE can be calculated by means of an algorithm that combines numerical integration with an optimization procedure that searches for an end condition [102].

To develop optimal bidding strategies for the GenCos in oligopolistic energy markets, Vahidinassab and Jadid study the impacts of GenCos' pollutant emission on their bidding strategies [6]. By neglecting demand-side bidding and bilateral contracts, the GenCos are assumed to submit quadratic bidding curves to the market and bid the quadratic coefficient of bidding curves under the locational marginal pricing (LMP) mechanism. The model employs SFE to represent each supplier's strategic behavior. Normal boundary intersection approach is used for generating Pareto optimal set and fuzzy decision making is employed to select the best compromise solution. The MCP is obtained based on the multi-objective optimal power flow method. The optimal bidding strategies are mathematically developed by using a bi-level optimization problem solution.

In view that most previous research mainly focuses on a single-period, single-market model, Gao and Sheble develop an SFE model for GenCos' bidding optimization in a multiple-period and multi-market scenario. For the proposed SFE model, they obtain the equilibrium condition by using discrete time optimal control which considers fuel resource constraints. The multiple-period optimal bid strategy is analytically derived by manipulating the intercept parameters of the SFE model. Both electricity and fuel markets are simultaneously considered, and the SFEs with resource constraint and transmission constraint are investigated, respectively [30]. Sahraei-Ardakani et al. propose an  $n$ -player dynamic game model to analyze the bidding strategy in an oligopolistic electricity market with either fixed or variable demands [103]. In this analytical short-term model, all players first choose their own strategy, and then, define an improvement matrix to improve the bidding strategy after seeing one another's strategy and the resulting payoff. A state-space approach is used to obtain the supply function equilibrium strategy for the model under many realistic constraints such as production bounds, ramping limits, up/down times, system contingencies, and transmission congestions.

Generally, in a centralized market, where the power pool collects suppliers' bids and loads and determines the dispatching schedule, the competition level as well as the model type is dependent upon the bidding procedure and the pricing rule. The GenCos may bid on prices without worrying about quantities

(Bertrand competition), but they may also bid on their production quantity as a function of the prices to be received from the equilibrium prices (SFE). In a decentralized market, the transactions are performed in bilateral or multilateral markets and the type of short-term competition is endogenous. The GenCos may compete by choosing the quantity they are willing to put on the market and the MCP is to be determined by an ISO (Cournot competition). The supply function model is a good compromise between Cournot and Bertrand competitions in a highly decentralized market [74].

#### 5.4. Other game theory based modeling methods

Besides the above three types of game based methods, some other game models are also proposed in recent literature [104–107,109,110]. They are listed in Table 7 and briefly discussed in the following.

Vytelingum et al. [104] first develop an adaptive-aggressiveness (AA) strategy which balances profit and transaction probability for participants in the Continuous Double Auction (CDA) power exchange market. Based on this, they further construct a two-population game model to analyze the participants' strategic bidding actions in the CDA market. Borghetti et al. [105] analyze the influence of GenCos' technical constraints on their bidding strategy selections using an optimization approach integrating the normal-form game theory (NGT) representation with a cost-based UC algorithm. The integration with the UC program enables the selection of the most convenient bidding strategy and embraces both an accurate estimation of production costs and various technical constraints. De la Torre et al. develop a Nash equilibrium (NE) based model for pool-based multi-period PoolCo markets under the assumptions of price-sensitive demands and profit-maximizing GenCos [106]. A model considering multi-period bidding, price elasticity, and network modeling of the market is built. Participants' bidding strategies can then be possibly detected by deriving Nash equilibria from the output data of an iterative simulation process. The strategies are iteratively evaluated to remove those dominated by others that generate higher profits [106]. Rodriguez and Anders present a methodology for a GenCo to design an hourly bidding strategy according to its degree of risk aversion (DRA) (either risk seeker or risk averse); the bidding curve is composed of optimal offer blocks determined from forecasted MCP and demand [107]. Bid Function Equilibria (BFE) is a model extended from multi-unit auction model [108] of wholesale electricity markets by assuming heterogeneous costs for different generation units of a GenCo. Crawford et al. investigate the application of BFE in the British electricity spot market and reveal that BFE is very effective for electricity market design and operation [109]. However, being different from SFE and other auction theories, BFE assumes the availability of complete information regarding each GenCo's marginal cost and the market demand, which is usually impractical. Liu et al. propose an incentive electricity bidding mechanism based

**Table 7**  
Other types of game theory based models for bidding strategy analysis.

| Ref.                     | Model bases | Auction rules | Markets        | Assumptions                                                                                 | Applications                                         |
|--------------------------|-------------|---------------|----------------|---------------------------------------------------------------------------------------------|------------------------------------------------------|
| Vytelingum et al. [104]  | AA          | UP            | General DAM    | Various market settings; CDA mechanism                                                      | Bidding in the CDA DAMs; market efficiency analysis  |
| Borghetti et al. [105]   | NGT         | UP            | General DAM    | Complete (but imperfect) knowledge of other participants' cost functions & strategies       | Bidding under feasibility constraints in DAMs        |
| De la Torre et al. [106] | NE          | UP            | PoolCo         | Price-sensitive demand                                                                      | Bidding in the multi-period PoolCo market.           |
| Rodriguez et al. [107]   | DRA         | UP            | Ontario PoolCo | Infinite fuel supply of fixed price                                                         | Thermal GenCo bidding in the pure PoolCo             |
| Crawford et al. [108]    | BFE         | UP            | E&W DAM        | Symmetric demand; no capacity limits; complete information                                  | Market design and bidding strategy analysis for DAMs |
| Liu et al. [110]         | GSM         | UP            | PoolCo         | Single price-quantity offer and bid; congestion, system ramp, and other constraints ignored | Market design; market power analysis                 |

on the signaling game theory, defined as the Generator Semi-randomized Matching (GSM) mechanism [110]. This new bidding mechanism is verified via a multi-agent simulation model. It is illustrated that, compared to the High-Low Matching (HLM) bidding mechanism, the GSM model can decrease the clearing price and GenCos' profits and increase the total transaction volume and the buyers' overall benefits.

## 6. Agent-based models

The restructured electricity markets typically involve price-quantity pairwise bids for the sale of large amounts of electricity by a small number of GenCos, resulting in extremely complex market processes in which traditional analytical and statistical tools are difficult to be applied [111]. In an agent-based model, market participants are modeled as adaptive agents with different bidding preferences and strategies, and the suppliers are enabled to utilize their past experiences to improve their behaviors in the market. Each agent may develop the optimal bidding strategy by learning from its past experiences obtained from the direct interaction with environment. This brings a new type of numerical analysis theory to deal with complex trading issues in the restructured electricity market [112]. Generally, the agent-based modeling procedure can be described as follows: (1) define the research questions to be resolved; (2) construct a model comprising an initial population of agents; (3) specify the initial model state by defining the agents' attributes and the structural and institutional framework of the electricity market within which the agents operate; (4) have the model evolve over time without further intervention; (5) analyze simulation results and evaluate the regularities observed in the data [113].

Agent-Based Computational Economics (ACE) is a fairly young research paradigm that offers methods for realistic electricity market modeling to overcome some shortcomings of the other methods discussed above [113]. A growing number of researchers have developed many agent-based models for simulating electricity markets [114,115]. However, compared with single GenCo optimization models and game theory based models, agent-based modeling studies for bidding strategy analysis are much less in literature. The representative publications are listed in Table 8.

Rahimiyan et al. compare the Q-learning (QL) approach [116] and the model-based (MB) approach in optimizing supplier's bidding strategy under electricity PAB auction rule [26]. The suppliers' behaviors are modeled in a multi-agent system, and the simulation results show that the Q-learning algorithm can enable the suppliers to find the optimal bidding strategy in the PoolCo market. For different PDFs, the QL algorithm can always converge to the optimal

solution obtained using the model-based approach. The results also show that the suppliers could adopt the bidding strategy according to their rivals' behavior and other effective factors in power system operation using the QL algorithm. Sheble proposes a genetic algorithm (GA) and an ACE framework for optimizing sellers' bidding strategies in a double-sided auction market where some players attempt to benefit from applying different strategies to cause economic instabilities and intentionally drive market prices [117]. Naghibi-Sistani et al. propose a Q-learning algorithm for the participants to find the optimal bidding strategy in the PoolCo electricity market. In this method, each bidder, independent of the others, learns about its state and the spot price. The results show that with the temperature variation reinforcement learning, the suppliers can learn the optimal policy found by game theory and can be adaptive to the parameter variations in the market [118].

Wen and David adopt Monte Carlo simulation and a refined genetic algorithm to build optimally coordinated bidding strategies for competitive suppliers in energy and spinning reserve markets. Under the assumptions that each supplier bids a linear supply function into the energy and spinning reserve markets, respectively, and that the two markets are dispatched separately, each supplier chooses the coefficients in the two linear supply functions to maximize the total benefits, considering the rivals' possible bidding policy [119]. They also apply a similar GA-based method to develop an overall bidding strategy for the suppliers in the day-ahead market [47]. Gountis and Bakirtzis propose a GA approach to optimize the profits of individual GenCos with multiple generating units. The model uses Monte Carlo simulation to calculate the expected profit and GA to find the optimal strategy. It is assumed that each supplier bids a linear supply function and considers the other bidders' bidding behaviors in the forms of PDF. However, such an assumption is usually not realistic since the rational generator can be expected in a competitive- hit81 bidding environment. Therefore, the profits estimated by the proposed algorithm are not realizable [120]. Earlier agent-based simulations employing GA algorithm for bidding strategy analysis can be found in [121,122].

Besides, some other learning algorithms have also been investigated. For example, Fujii et al. apply a multi-agent model, which learns a bidding strategy autonomously through trial-and-error search action, to numerically analyze the price formation process of an open electricity market [123]. The model is believed to be helpful in analyzing more general electricity markets which have several different types of power plants with unit commitment costs, seasonal and hourly demand fluctuation, real-time regulation market and operating reserve market. Bunn et al. develop an agent-based simulation model by using computational learning (CL) algorithm to investigate the impact of vertical integration between

**Table 8**  
Representative agent-based modeling methods for bidding strategy analysis.

| Ref.                         | Learning rules | Auction rules  | Markets | Assumptions                                                                                                         | Applications                                                                      |
|------------------------------|----------------|----------------|---------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Rahimiyan et al. [26]        | MB             | PAB            | PoolCo  | Stochastic MCP of known PDF                                                                                         | Adaptive bidding in the deregulated PoolCo                                        |
| Rahimiyan et al. [26]        | QL             | PAB            | PoolCo  | Past experiences learned to find optimal bid price                                                                  | Adaptive bidding in the deregulated PoolCo                                        |
| Sheble [117]                 | GA             | UP             | PoolCo  | Double-sided auction                                                                                                | Market power analysis in double-sided auction                                     |
| Naghibi-Sistani et al. [118] | QL             | UP             | PoolCo  | Unknown & unnecessary rivals' information                                                                           | Adaptive bidding in the deregulated PoolCo                                        |
| Wen et al. [119]             | GA             | UP             | Hybrid  | Linear SF of energy & spinning reserve markets being separately dispatched                                          | Adaptive bidding in the deregulated PoolCo including spinning reserve markets     |
| Gountis et al. [120]         | GA             | UP             | PoolCo  | Linear supply function; known rivals' bids                                                                          | Adaptive bidding in the deregulated PoolCo                                        |
| Fujii et al. [123]           | TAE            | UP             | Hybrid  | Different types of GenCos; seasonal & hourly demand fluctuation; real-time regulation and operating reserve markets | Adaptive bidding in the deregulated PoolCo & reserve markets                      |
| Bunn et al. [124]            | CL/MA          | PAB            | PoolCo  | Price-taking demand, GenCos offering above marginal cost; capacity payments ignored                                 | Adaptive bidding in the deregulated PoolCo including Korean market; Market design |
| Azadeh et al. [125]          | ACO            | UP/PAB/Vickrey | PoolCo  | Fixed demand; no transmission constraint; a step-wise discrete bid function.                                        | Adaptive bidding in the deregulated PoolCo; design of market clearing rule        |

electricity generators and retailers on market power in a competitive wholesale market setting [124]. It is found that in various cases, whilst vertical integration generally reduces spot prices, it can increase or decrease the market power of other generators, depending upon the market share and the technology segment of the market.

Azadeh et al. propose an agent-based simulation model based on Ant Colony Optimization (ACO) algorithm to compare three wholesale electricity market clearing strategies: UP, PAB, and generalized Vickrey rules. The proposed model is suitable for high-dimension bidding functions and enables modelers to avoid “curse of dimensionality”. They investigate step-wise discrete bid functions and their impacts on the efficiency of the market under different available market settlement rules. The assumptions in the proposed algorithm include: inelastic and fixed demand; call market with different price settlement rules (UP, PAB, and generalized Vickrey rules); no transmission constraint; and a step-wise discrete bid function. The method can solve dynamic and static combinatorial optimization problems of market strategy optimization [125].

Besides, there are a few other studies on agent-based modeling applications in literature [25,126–132]. Xiong et al. compare the UP and PAB auction rules by using a multi-agent approach, where each adaptive GenCo develops the bidding prices based on Q-learning algorithm [25]. The experimental results show that the PAB auction can result in lower expected market prices and price volatility. It is also shown that the demand-side response has less effect in reducing market prices under the PAB auction rule since in this auction the bidders bid as close to the market prices as possible, which makes the aggregate supply curve more flattened than that under the UP rule. Xiong et al. propose an evolutionary daily bidding strategy for the supplier in a perfectly competitive day-ahead electricity auction market under the PAB auction rule [126]. The feasibility of the proposed bidding strategy is verified by agent-based simulations. Walter and Gomide present a GA-based approach to obtain evolutionary GenCo bidding strategies in the power market. Simulation results illustrate that the strategies derived by this approach is superior in enhancing the GenCo's profitability over the commonly adopted marginal cost-based approaches [127]. They also introduce a co-evolutionary algorithm to obtain a profitable bidding strategy for the participant by using information commonly available in a much dynamic environment. The results demonstrate that this approach can further improve the GenCo's profits [128]. Gao et al. build an adaptive multi-agent model to analyze and compare the application performances of genetic algorithm, evolutionary programming, and particle swarm in simulating participants' bidding behaviors [129]. Sueyoshi and Tadiparthi develop an agent-based decision support system (DSS) for analyzing the dynamic price change in the competitive electricity wholesale environment. The proposed DSS is effective in assessing new trading strategies in the electricity PoolCo market [130].

Agent-based models can mimic human behaviors and simulate optimal bidding procedures. In such models, market participants are handled as adaptive agents with different bidding preferences and strategies, and their bidding decisions are influenced by many uncontrollable factors. This group of models is more flexible, robust, and easily implemented compared with the previous two groups of approaches, and thus it opens up a new type of modeling analysis to deal with business complexity of electricity trading. However, the underlying mathematical foundation for agent-based modeling has not yet been clearly verified to date [30].

## 7. Hybrid and other modeling methods

Besides the above three major groups of modeling approaches, there are a few other innovative methods developed recently for

strategic bidding analysis. In particular, the hybrid approach that combines multiple modeling methods stimulates enormous interests among the researchers [133–136].

Yamin and Shahidepour develop a hybrid model combining LR algorithm and GA for the GenCos to generate a proper unit commitment scheduling and derive the optimal supply curves. It is demonstrated that the proposed hybrid model is better than the LR approach and the traditional unit commitment approach in terms of helping the GenCos to increase profits [133]. They also adopt the augmented Lagrangian relaxation algorithm to solve self-scheduling and energy bidding problems in competitive electricity markets constrained by transmission congestions, fuel, and emission. The supply curve is derived as a function of generation schedule to achieve the maximum profit. The slope of the supply curve is dependent upon the forecast price and the power output obtained in the self-scheduling result [134].

Sueyoshi proposes an agent-based approach equipped with game theory for analyzing the strategic collaboration among learning agents during a dynamic market change in the 2000–2001 California electricity crisis [135]. The concept of partial reinforcement learning is incorporated into trading agents who can learn from both the dynamic market change and the collaboration with other traders. It is found that the learning speed of traders becomes slow when a large fluctuation occurs in the power exchange market. Azevedo and Correia propose a model by combining Bayes' rule and the game theory for the participants in the first stage of an electric energy bilateral contract auction in the Brazilian market. The model can help one agent to attribute bids to the other agents and observe the consequences [136].

In addition, Song et al. present the concept of conjectural variation (CV) and its applications to the strategic bidding of GenCo in the oligopolistic electricity spot market. The conjecture of a firm is defined as its belief or expectation of how its rivals will react to the change of its output. It is verified that, in a real spot market containing multiple players, CV based bidding strategy (CVBS) enables a GenCo to integrate its rivals' responses into one pseudo-competitor's response and to make optimal decision accordingly based on available imperfect information announced in the market. It is also demonstrated that classical game theoretical bidding strategies (GTBS), such as Bertrand, Cournot, Stackelberg and monopoly (collusion), are actually the special cases in the CVBS solution family [29].

## 8. Conclusive remarks

In the near future, GenCos are encouraged to directly participate in the liberalized electricity wholesale market by submitting their strategic supply offers. This makes it possible for the GenCos to maximize their profits by optimizing their bidding strategies. In the literature, many models have been proposed to analyze the bidding behaviors in the oligopolistic electricity wholesale market and their performances depend on different market structures, auction mechanisms, and other factors. This paper provides a systematical review on the recent publications analyzing the bidding strategy in the liberalized electricity wholesale market. Generally, various models can be classified into three main groups: single GenCo optimization models, game theory models, and agent-based models. Besides, some hybrid models have also been developed recently.

In addition to the dynamics and uncertainty of the load, the price, and the renewable generation, the GenCos bidding strategy is directly affected by and dependent upon the deregulated electricity market itself. However, since the electricity wholesale market is still under reconstruction, many interesting and meaningful issues related to bidding optimization under various possible market designs should be investigated and tackled, which include but are not limited to the following topics:

- Bidding under different market prototypes: Various electricity market structures are being designed and tested to ensure free access, fair competition, high efficiency, and systems security and reliability. The analysis on GenCos' strategic bidding or market power is important for the market design efforts.
- Bidding under PAB auction rule: Bidding strategy under the discriminatory pricing rule has its advantages and disadvantages in deregulating the market. However, the relevant research efforts are still far from enough.
- Bidding under double-sided auction mechanism: Allowing the buyers as well as the sellers to submit their competitive bids, double-sided auctions are commonly regarded as a better setting for deregulating the electricity markets. However, bidding under such settings may also face with increased uncertainty and risks, and this calls for quantitative modeling analysis.
- Bidding in hybrid or combined markets: The hybrid market often exists in which electrical energy is traded together with spinning reserves simultaneously or in the presence of future contracts and bilateral contracts. Also, with the combination of day-ahead, hour-ahead, and real-time markets, many reconstructed electricity markets allow the participants to update day-ahead supply offers and purchasing bids before the actual delivery. Besides, the future electricity markets, by combining the wholesale and retailing markets, should allow the end users to bid in the markets directly. Bidding optimization in the combined markets will be a significant topic for research.
- Advance in modeling methods and algorithms: To better represent the transaction behaviors in complex electricity markets, the shortcomings of each method should be overcome by continuously pushing for new theoretical developments.
- Bidding cooperatively or with internal control strategy: Without affecting the oligopolistic nature of the deregulated electricity market, locational GenCos, especially for those owning different generation resources, could bid in the market cooperatively. For example, a WGenCo can cooperate with neighboring WGenCos or other type of GenCos, or the WGenCo can adopt some internal balancing strategies, e.g., integrating the wind generation with storage technology. The effects of such cooperative bidding strategies or internal controls should be further investigated.
- Risk management under uncertainty: To deal with the risks of bidding under the uncertainty in demand, price and production, different risk control measures and operational scenarios need to be investigated. This is especially important for those renewable GenCos with distributed and intermittent productions. For instance, besides the uncertain load and electricity price, the wind generators should also consider the uncertain wind generations before submitting their supply offers to the electricity markets.

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